1 Change 800 millilitres to litres

0.8 litres

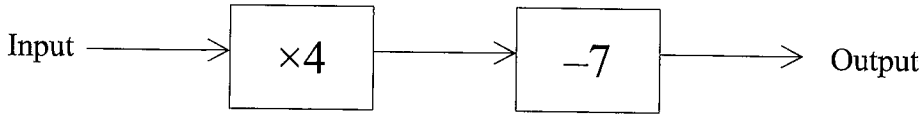
(Total for Question 1 is 1 mark)

2 Change 50 kilograms to grams.

50 000 grams

(Total for question 2 is 1 mark)

3 Here is a number machine.



(a) What is the **output** when the **input** is 5?

$5 \times 4 = 20$

$20 - 7 = 13$

13

(1)

(b) What is the **input** when the **output** is 29?

$29 + 7 = 36$

$36 \div 4 = 9$

1 mark for 29+7 or 36

9

(2)

(Total for Question 3 is 3 marks)

4 Which is greater

24% of 140 or 31% of 110

You must show your working.

$\frac{140}{10} = 14$ (10%)

$\frac{140}{100} = 1.4$ (1%)

$1.4 \times 4 = 5.6$ (4%)

$\begin{array}{r} 14 \\ 14 \\ 5.6 \\ \hline 33.6 \end{array}$

1 mark for 33.6

$\frac{110}{10} = 11$ (10%)

$\frac{110}{100} = 1.1$ (1%)

$11 \times 3 = 33$ (30%)

$33 + 1.1 = \underline{34.1}$ (31%)

1 mark for 34.1

31% of 110

All 3 marks for 33.6, 34.1 and correct answer

(Total for Question 4 is 3 marks)

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5 (a) Write the ratio 25 : 40 in its simplest form.

$\div 5 \quad \div 5$

5 : 8

5 : 8

(1)

(b) There are red shapes and blue shapes in a box, $\frac{5}{8}$ of the shapes are red.

Write the ratio of red shapes to blue shapes.

$\frac{5}{8} : \frac{3}{8}$

5 : 3

5 : 3

(1)

(Total for Question 5 is 2 marks)

6 The accurate scale drawing shows two towns, Town A and Town B.

Town A
X

X Town B

The scale is 1:50000

Find the real distance between Town A and Town B, in kilometres.

5cm 1 mark for 5cm

5 × 50 000 = 250 000 cm 1 mark for fully correct method

= 2500 m

= 2.5 km

2.5 km

(Total for Question 6 is 3 marks)

---

7 Stan has an income of £3000 a month.

He spends $\frac{2}{5}$ of his income on rent. $\frac{3000}{5} = 600$ $2 \times 600 = 1200$

He spends $\frac{4}{15}$ of his income on bills. $\frac{3000}{15} = 200$ $4 \times 200 = 800$

He spends $\frac{1}{10}$ of his income on food. $\frac{3000}{10} = 300$

Stan saves the rest of his income.

Work out how much Stan saves each month. 1 mark for 1200, 800 or 300

Stan spends ~~600 + 200 + 300 = £1100~~

1200 + 800 + 300 = £2300

Stan saves 3000 − 2300 = £700

1 mark for 2300 or just one mistake

£ 700

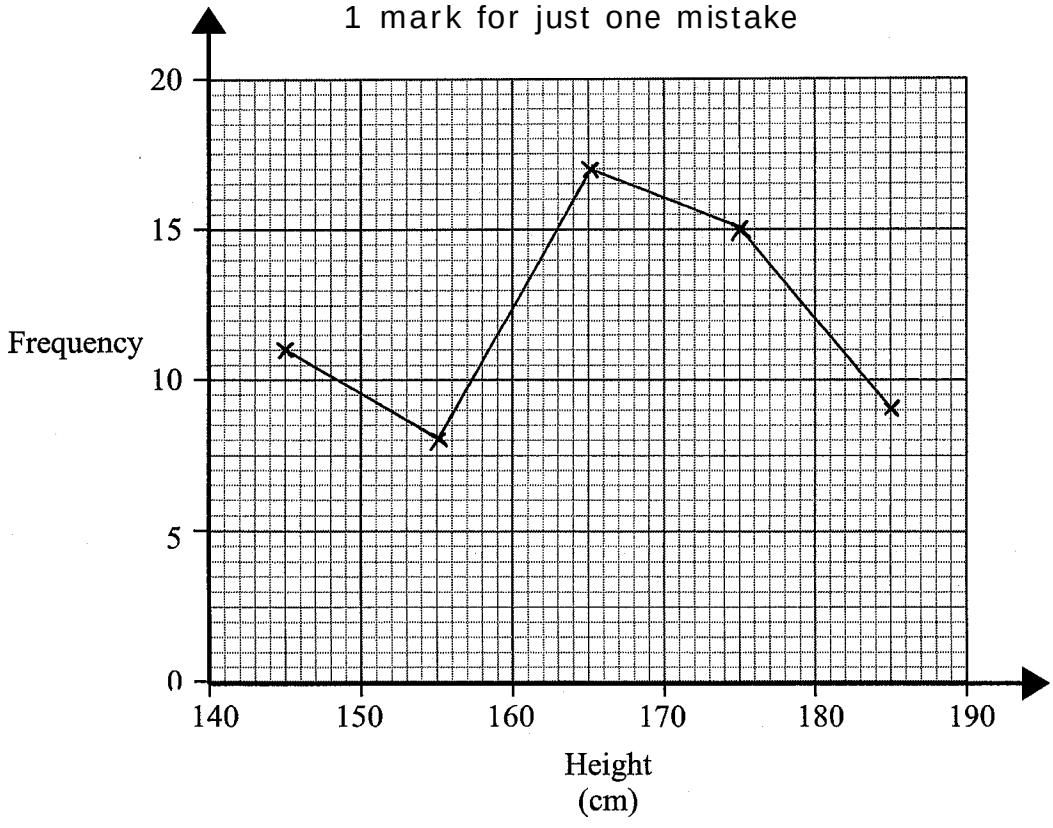
(Total for question 7 is 3 marks)

8 The frequency table shows the heights, in cm, of some tomato plants.

Height (cm)	Frequency
$140 < h \leqslant 150$	11
$150 < h \leqslant 160$	8
$160 < h \leqslant 170$	17
$170 < h \leqslant 180$	15
$180 < h \leqslant 190$	9

Draw a frequency polygon to show this information.

1 mark for just one mistake



(Total for Question 8 is 2 marks)

9 (a) Work out $2\frac{1}{3} + 1\frac{2}{5}$

$$\frac{5\times 7}{5\times 3} + \frac{7\times 3}{5\times 3}$$

$$\frac{35}{15} + \frac{21}{15} = \frac{56}{15} = 3\frac{11}{15}$$

1 mark for same denominators

$3\frac{11}{15}$

(2)

(b) Work out $2\frac{1}{4} \div \frac{3}{5}$

Give your answer as a mixed number in its simplest form.

$$\frac{9}{4} \div \frac{3}{5}$$

$$\frac{9}{4} \times \frac{5}{3} = \frac{45}{12} = 3\frac{9}{12} = 3\frac{3}{4}$$

1 mark for 45/12 or equivalent

$3\frac{3}{4}$

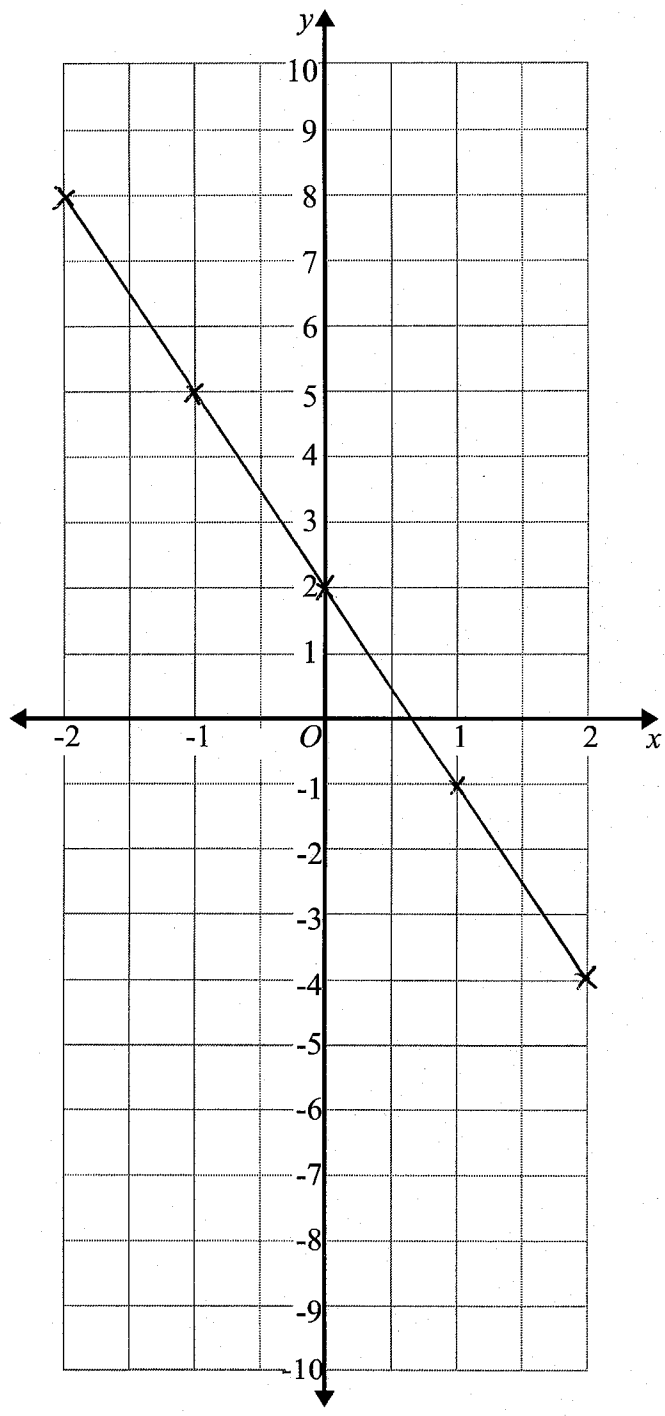
(2)

(Total for Question 9 is 4 marks)

10 On the grid, draw the graph of $y = 2 - 3x$ for values of x from -2 to 2

x	-2	-1	0	1	2
y	8	5	2	-1	-4

1 mark for any correct coordinates/table



1 mark for just one mistake

(Total for Question 10 is 3 marks)